

Large language model-driven decision-making for single-atom Fenton-like catalysts toward wastewater treatment

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HIGHLIGHTS

- LLM framework processes 1000+ papers with >87 % extraction accuracy.
- Few-shot CoT prompting enables high-fidelity scientific knowledge extraction.
- Structured knowledge reveals key synthesis-structure-performance relationships.
- ML platform guides catalyst selection based on target wastewater conditions.
- Experiments validate platform-recommended catalysts achieve predicted performance.

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ABSTRACT

Tailoring catalysts to diverse environmental conditions for wastewater treatment demands rational material design, yet such decisions remain largely empirical due to fragmented literature knowledge. Here we establish an LLM-powered framework designed to underpin data-driven decision-making in Fenton catalysis. Using optimized few-shot Chain-of-Thought prompting, our framework processed over 1000 domain publications and patents on Fenton-like single-atom catalysts (SACs), systematically extracting knowledge including synthesis, properties, and performance metrics with over 87 % accuracy and 90 % completeness. This process generated a high-fidelity knowledge baseline, revealing key synthesis-structure-property relationships governing SAC performance. Building on these structured insights, we developed an interactive decision-support platform that integrates machine learning-based performance prediction with literature-anchored synthesis recommendations, guiding researchers toward optimal catalyst configurations based on reaction condition. Experimental validation confirmed that platform-recommended catalysts achieved predicted performance under realistic conditions. This work demonstrates a transferable paradigm for intelligent decision-making, accelerating the development of robust catalysts for advanced water purification.

Global environmental crises pose escalating threats to human health and sustainable development (Hodges et al., 2018; Schwarzenbach et al., 2006; Shi et al., 2025; Vogt and Weckhuysen, 2022). Among these, water contamination by organic pollutants has emerged as one of the most pressing challenges (Rockström et al., 2023). These contaminants, ranging from industrial dyes and phenolic compounds to pharmaceutical residues, often resist conventional biological treatment due to their

structural complexity and chemical stability (Persson et al., 2022; Wilkinson et al., 2022). Compounding this difficulty is the diversity of real-world treatment scenarios, which span industrial effluents, contaminated groundwater, and municipal wastewater, each presenting distinct pH conditions, coexisting species, and operational constraints (Yu et al., 2024b). The heterogeneity in both pollutant characteristics and environmental matrices demands remediation technologies with

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tunable performance that can be rationally adapted to specific application scenarios (Mauter et al., 2018).

Among various remediation strategies, advanced oxidation processes (AOPs) based on Fenton-like chemistry have emerged as a powerful approach for degrading recalcitrant pollutants through the generation of highly reactive oxygen species (Shang et al., 2021). Within this paradigm, single-atom catalysts (SACs) represent a transformative class of materials characterized by atomically dispersed metal centers, unique electronic configurations, and precisely engineered coordination environments (Cui et al., 2018; Ji et al., 2020; Shang et al., 2021). These structural features confer exceptional catalytic selectivity and intrinsic activity while maximizing atom utilization efficiency, establishing SACs as a promising frontier for wastewater treatment (Chen et al., 2025; Jin et al., 2025; Zhang et al., 2025b). However, the performance of SACs is highly sensitive to the interplay between synthesis parameters, support chemistry, metal coordination, and reaction condition (Li et al., 2024; Xiong et al., 2022; Yang et al., 2025; Zhang et al., 2025a), resulting in a vast and multidimensional design space. Rational decision-making that matches specific SAC properties to target application scenarios is therefore essential, yet remains a central challenge in Fenton catalysis.

Efforts to address this challenge have primarily relied on individual experimental studies and accumulated empirical knowledge (Gu et al., 2023; Huang et al., 2024). These investigations, while crucial for uncovering fundamental mechanisms and screening promising SAC candidates, typically concentrate on narrow material properties or constrained reaction conditions, producing fragmented understanding insufficient for systematic decision-making (Zheng et al., 2025a). Traditional approaches to consolidating knowledge have employed manual literature surveys and meta-analytical frameworks (Silva et al., 2025; Wu et al., 2023; Zhang et al., 2022). Through these methods, important discoveries have emerged regarding the role of specific metal sites in radical generation pathways and of support materials in influencing catalyst stability and activation efficiency (Rodríguez-Padrón et al., 2019; Shang et al., 2021; Yang et al., 2022). Nevertheless, these manual processes require months of specialized effort and yield static conclusions that cannot readily adapt to new pollutants or emerging application scenarios. The absence of systematic knowledge frameworks that can inform rational decision-making in SAC design has become a persistent bottleneck.

Recent breakthroughs in Large Language Models (LLMs) offer a paradigm shift in how we extract, organize, and synthesize scientific knowledge (Bran et al., 2024; Zeni et al., 2025; Zheng et al., 2025b). LLMs possess a remarkable capacity to parse unstructured text, enabling them to identify salient entities, decipher complex scientific arguments, and systematically organize extracted information to support knowledge-driven decision-making (Dagdelen et al., 2024; Yu et al., 2024a; Zheng et al., 2023). Herein, we develop a comprehensive LLM-based framework to the intelligent decision-making of Fenton-like SAC design. Using a curated corpus of over 1000 domain publications, our framework systematically extracts synthesis conditions, structural and electronic properties, and catalytic performance metrics with high accuracy. Statistical analysis of the extracted data uncovers key synthesis-structure-performance relationships. Building upon this structured knowledge base, we establish an interactive decision-support platform assisted by machine learning (ML) that guides researchers in catalyst selection and optimization based on target pollutants, reaction conditions, and performance requirements. We expect that the proposed framework will accelerate rational catalyst design and serve as a transferable paradigm for intelligent decision-making across diverse wastewater treatment scenarios.

1. Materials and methods

1.1. Corpus development

The foundational corpus was constructed through a systematic

literature search of Web of Science, Scopus, Google Scholar, and key patent repositories (USPTO, EPO, WIPO) for publications between 2017 and 2025. Search queries combined keywords for single-atom catalysts, Fenton-like reactions, and organic pollutants, refined using Boolean operators. A rigorous, multi-stage curation process, including expert validation and citation analysis, yielded a final corpus of 932 peer-reviewed articles and 135 patents. The full methodology for corpus development, including detailed search queries and inclusion/exclusion criteria, is provided in **Supplementary Note S1**.

1.2. LLM selection and prompt engineering

A comparative analysis of LLMs, including GPT-4o, GPT-3.5-turbo, LLaMA-2, Vicuna, and Alpaca, was conducted to select the optimal model based on F1-score for scientific data extraction. GPT-4o was selected for its superior performance. We systematically designed and evaluated zero-shot, few-shot, and few-shot Chain-of-Thought (CoT) prompting strategies to extract information on synthesis, properties, and performance. The few-shot CoT strategy, which integrates explicit reasoning steps, was identified as the optimal approach for building the knowledge baseline. The extraction pipeline processes the full text, embedded tables, and available supplementary materials of each publication in machine-readable format, capturing the quantitative parameters that are conventionally reported as numerical values in the Fenton-like SAC literature. A detailed description of the LLM evaluation, prompt design, and strategy assessment is available in **Supplementary Notes S1 and S2**.

1.3. Input feature selection and preprocessing for ML

For ML modeling, a core set of eight features with high data coverage was selected: support material, central metal atom, metal loading, coordinating atom, coordination number, metal oxidation state, specific surface area, and binding energy. Limited missing values were imputed using a k-nearest neighbors (KNN) algorithm, with the hyperparameter $k = 5$ optimized via grid search. Categorical features were converted into numerical format using one-hot encoding. This preprocessing resulted in 14 catalyst descriptor features. Together with 8 environmental parameter features introduced during ML modeling, the complete input comprised 22 features. An interquartile range (IQR)-based outlier detection step was further applied to the standardized k distribution to flag and verify potential data conflicts or extraction errors. The complete feature selection, imputation validation, and preprocessing workflow is detailed in **Supplementary Note S1**.

1.4. ML modeling and interpretation

A suite of ML algorithms was selected to comprehensively explore potential linear and complex non-linear relationships between the 22 input features and the catalytic performance (standardized k value, min^{-1}). The chosen models included Random Forest, XGBoost, LightGBM, and LASSO regression (detailed rationale for model selection is provided in the **Supplementary Note S3**) (Breiman, 2001; Chen and Guestrin, 2016; Ke et al., 2017). The dataset was partitioned into training (80 %) and testing (20 %) sets, with stratification implemented based on the activator type (PMS or H_2O_2). To strictly prevent data leakage, all preprocessing steps (KNN imputation, one-hot encoding, and feature standardization) were fitted exclusively on the training fold and then applied to the test fold within each cross-validation iteration. The predictive performance of each trained model was rigorously evaluated using standard statistical metrics: the coefficient of determination (R^2), mean squared error (MSE), mean absolute error (MAE), and Pearson correlation coefficient (r).

To gain insights into the predictive mechanisms of the models and to quantify the relative importance of different input features in determining catalytic performance, SHAP (SHapley Additive exPlanations)

analysis was employed (Hancock et al., 2025). This technique provides a principled way to attribute the model's predictions to its input features (Wang et al., 2024). For categorical variables that had undergone one-hot encoding, the SHAP values corresponding to the derived set of binary features were aggregated by summing their absolute values to reflect the overall contribution of the original categorical variable (Jullum et al., 2021). Global feature importance rankings were subsequently determined from the average absolute SHAP values for both PMS and H_2O_2 activation systems.

1.5. Catalyst synthesis and experimental validation

To validate the reliability of platform recommendations, Fe-based SACs were synthesized following the protocols recommended by the

decision-support platform. For PMS activation systems, the platform recommended $\text{Fe-N}_3/\text{C}$ configuration, which was synthesized via pyrolysis of iron phthalocyanine with polyaniline/melamine at 900°C under NH_3 atmosphere. For H_2O_2 activation systems, the platform recommended $\text{Fe-N}_5/\text{C}$ configuration, which was synthesized via pyrolysis of iron phthalocyanine with polyaniline/melamine at 500°C under Ar atmosphere. Coordination numbers were confirmed using XAS. The Fenton-like catalytic performance was evaluated through the degradation of phenol (20 mg/L) using both PMS and H_2O_2 as activators. Reaction kinetics were monitored using high-performance liquid chromatography (HPLC). Further experimental details are in **Supplementary Note S1**.

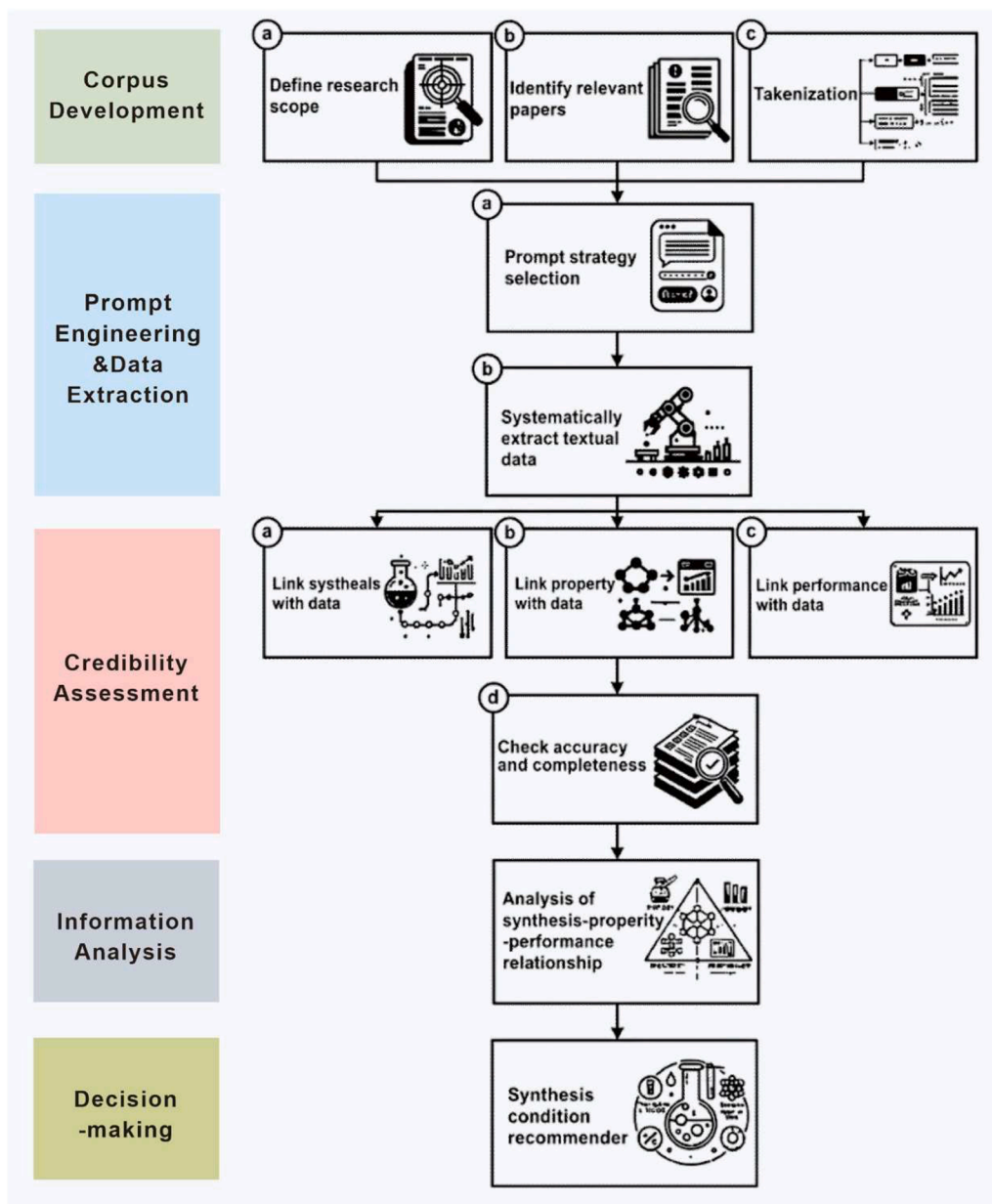


Fig. 1. LLM-driven framework for intelligent decision-making of single-atom Fenton-like catalysts for wastewater treatment. The workflow begins with Corpus Development (green section), encompassing research scope definition, relevant paper identification, and document tokenization. Prompt Engineering and Data Extraction (blue section) follows, where optimized prompting strategies systematically extract textual data linking synthesis, properties, and performance metrics. Credibility Assessment (red section) validates extraction accuracy and completeness. Information Analysis (gray section) uncovers synthesis-structure-property relationships from the validated knowledge baseline. The framework culminates in Decision-making (yellow section), where an interactive platform recommends optimal catalyst configurations and synthesis conditions for targeted wastewater treatment.

2. Results and discussion

2.1. Overview of the LLM-based framework

The LLM-based framework established in this study operates through a systematic workflow designed to transform fragmented literature knowledge into an integrated decision-support system for rational catalyst design, as illustrated in Fig. 1. With Fenton-like single-atom catalysis serving as a proof-of-concept, the workflow progresses through five interconnected stages: corpus development, prompt engineering and data extraction, credibility assessment, information analysis, and decision-making.

The process initiates with corpus development (green section), where systematic literature searches and rigorous curation yielded a high-quality repository. The framework then advances to prompt engineering and data extraction (blue section), where optimized prompting strategies guide the LLM to systematically extract synthesis parameters, structural properties, and performance metrics from the curated corpus. The extracted data undergoes credibility assessment (red section) to verify accuracy and completeness against source publications before inclusion in the final dataset. The validated dataset enables information analysis (gray section), where statistical methodologies reveal synthesis-property correlations while application-oriented analyses examine how reaction conditions impose distinct requirements on catalyst design. The framework culminates in decision-making (yellow section), featuring an ML-driven platform where users input application requirements and the system identifies optimal catalyst configurations with corresponding synthesis protocols. Experimental validation confirmed the reliability of platform recommendations under realistic conditions.

2.2. Prompt engineering for knowledge baseline construction

Our framework is built upon a high-quality core corpus comprising comprehensive scientific knowledge on Fenton-like SACs. This corpus was constructed through a multi-stage process (detailed in **Supplementary Note S1**), beginning with extensive keyword searches in prominent scientific databases and patent repositories. Search queries combined specific terms for SACs, Fenton-like reactions, and relevant pollutants, refined by Boolean operators for optimal specificity and recall. Strict inclusion criteria prioritized peer-reviewed articles and patents from journals published between 2017 and 2025 that reported experimental data on SAC synthesis, characterization, and Fenton-like performance. The relevance and quality of each potential publication were then validated by at least two independent domain experts, with discrepancies adjudicated by a third reviewer. This rigorous curation finally yielded a core corpus of 932 peer-reviewed publications and 135 patents, establishing a comprehensive knowledge foundation that captures the full spectrum of synthesis approaches, material configurations, and application scenarios essential for subsequent decision support.

With the corpus established, we proceeded to extract targeted information through prompt engineering. We designed the extraction protocol to collect three categories of information: (1) detailed catalyst synthesis methods and parameters; (2) precise descriptions of structural and electronic properties; and (3) comprehensive data on Fenton-like performance. Three prompting methodologies (zero-shot, few-shot, and few-shot CoT) were selected for systematic evaluation and comparison. To mitigate potential hallucinations and enhance extraction reliability, we incorporated combined strategies including designing clear instructions and specifying desired output formats, selecting the optimal LLM through comparison (favoring GPT-4o), and validating the extracted information against predefined lists. A more detailed explanation of the mitigation strategies can be found in **Supplementary Note S1**.

We focused initially on extracting structural and electronic properties of SACs to determine the optimal prompting approach. These properties were selected as evaluation criteria for two reasons: first, they

are crucial determinants of catalytic performance directly controlled by synthesis parameters and thus central to rational catalyst design; second, their extraction is inherently challenging as they are often embedded implicitly within synthesis and characterization descriptions using varied nomenclature and reporting formats. The evaluation revealed significant performance differences among the three methodologies. Zero-shot prompting provides the LLM with direct instructions without examples (Wei et al., 2021). When applied systematically to our corpus using the extraction instructions in **Table S1**, this approach achieved low accuracy (42 %) and recall (51 %) scores, indicating frequent failures to capture crucial data relationships (Fig. 2a). The model struggled with complex scientific terminology and frequently produced inaccurate property associations, such as incorrectly assigning multiple coordination environments to a single material when each actually corresponded to distinct catalysts. Few-shot prompting markedly improved extraction performance by leveraging the model's in-context learning capabilities (Cahyawijaya et al., 2024), achieving 72 % accuracy, 70 % recall, and 61 % F1 Score while substantially reducing hallucination rates (Fig. 2a, b). However, performance degraded when handling feature categories or levels of specificity not represented in the example set, particularly for complex coordination environments not covered by training examples.

The few-shot CoT prompting strategy represents a significant advancement by incorporating explicit reasoning steps within provided examples, building upon the in-context learning capabilities of few-shot prompting while enabling more sophisticated information processing (Kojima et al., 2022; Yu et al., 2023). We designed domain-specific CoT prompts that guide the model through analyzing catalyst properties in a sequential manner mimicking the thought process of domain experts (**Table S1**). Results showed that few-shot CoT prompting achieves consistently high-performance metrics across our corpus, with F1-scores exceeding 90 % for all 12 key properties extracted (Fig. S1). Notably, parameters such as coordination geometry that could not be reliably recognized by few-shot prompting were successfully identified through the CoT approach (Fig. S1, Fig. S2). The extracted property landscape revealed that carbon-based matrices (~60 %), metal-organic frameworks (~30 %), and oxides (~10 %) constitute the predominant supporting materials, while the coordination sphere is predominantly characterized by N atoms (~95 %) and O atoms (~20 %) in stabilized M-N or M-O structural motifs (Fig. 2d). Regarding the central metal, Fe dominates the corpus (~40 %), followed by Co and Cu, with these three transition metals accounting for approximately 95 % of reported SAC systems, reflecting their superior catalytic performance and cost-effectiveness for Fenton-like processes.

Subsequently, we applied the optimal few-shot CoT method to systematically gather information on SAC synthesis and Fenton-like performance. For the extraction of synthesis information, a specialized few-shot CoT prompting template was designed and validated across comprehensive synthesis parameters (**Table S1**). This validation yielded excellent precision (>90 %), recall (>87 %), and F1-scores (>82 %) for all evaluated parameters, including synthesis method, metal precursors, solvent conditions, reaction temperature and time, treatment atmosphere, and modulators (Fig. 2c). Recognizing that Fenton-like reactions depend critically on activator selection, a specialized prompt was designed to identify activator types and extract associated performance data (**Table S1**). Our extraction process categorized activators into PMS (64.3 %), H₂O₂ (28.6 %), and other activators (7.1 %) (Fig. S3). The central performance metric targeted was the reaction rate constant (k , min⁻¹), with prompts designed to simultaneously extract key reaction parameters enabling standardization to a common unit for cross-study comparison (detailed in **Supplementary Note S1**). The extraction achieved precision, recall, and F1-scores exceeding 90 % across all activator systems (Fig. 2e). Through this systematic and accurate extraction of synthesis procedures, material properties, and standardized performance metrics, we established a high-fidelity knowledge baseline ready for subsequent analysis and decision support. This knowledge baseline, comprising validated data entries across diverse

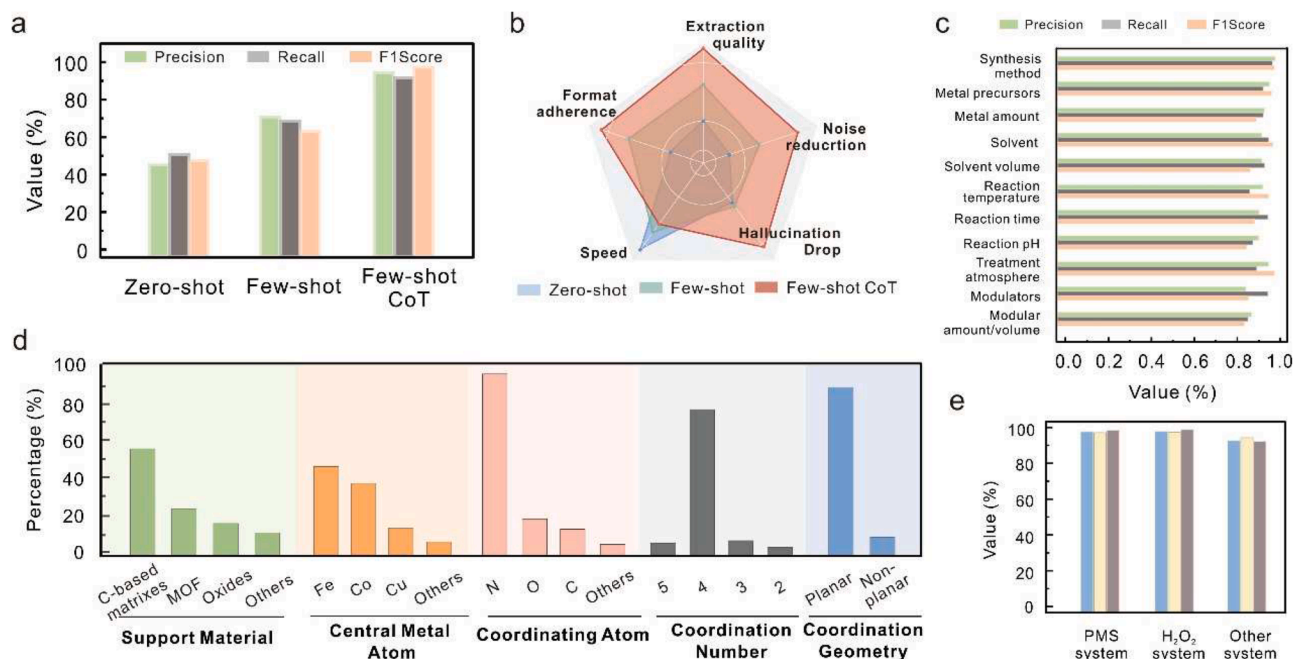


Fig. 2. Prompt engineering optimization and knowledge extraction validation. (a) Performance comparison of zero-shot, few-shot, and few-shot CoT prompting strategies across precision, recall, and F1-score metrics. (b) Radar chart evaluation of prompting strategies across six dimensions including extraction quality, noise reduction, hallucination drop, speed, format adherence, and overall performance. (c) Extraction accuracy for synthesis parameters demonstrating >82 % precision, recall, and F1-score across all evaluated categories. (d) Distribution of extracted catalyst properties including support material, central metal atom, coordinating atom, coordination number, and coordination geometry from the literature corpus. (e) Extraction performance validation across PMS, H₂O₂, and other activator systems.

catalyst configurations and reaction conditions, provides the quantitative foundation for uncovering the synthesis-structure-

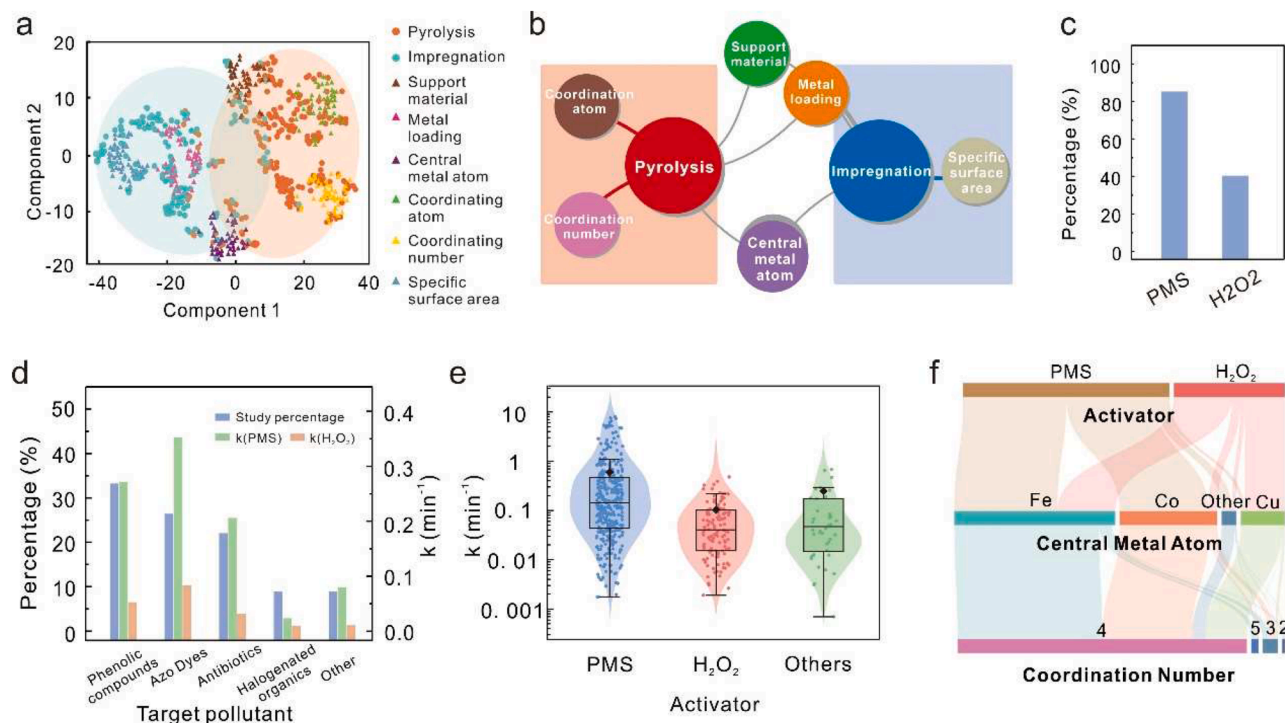


Fig. 3. Statistical analysis of synthesis-structure-property relationships. (a) t-SNE visualization revealing distinct property domains for pyrolysis and impregnation synthesis methodologies. (b) Schematic illustration of synthesis method-property correlations, showing pyrolysis emphasizes coordination environment while impregnation focuses on metal loading and surface area. (c) Percentage of reported SACs retaining high catalytic activity under neutral-to-alkaline conditions for PMS and H₂O₂ activation systems. (d) Target pollutant distribution and corresponding reaction rate constants (k) for PMS and H₂O₂ systems. (e) Violin plot comparison of catalytic performance across different activator systems. (f) Sankey diagram illustrating the conditional dependencies among activator type, central metal atom, and coordination number.

performance relationships that will inform our decision-support platform. We next analyze these relationships to identify the key design principles governing SAC performance under different application scenarios.

2.3. From knowledge baseline to synthesis-property-performance insights

Drawing from the established knowledge baseline, we analyzed synthesis-property relationships and application-dependent performance patterns to establish the inferential foundation for subsequent decision support. Our investigation identified pyrolysis and impregnation as the predominant SAC synthesis techniques, collectively accounting for 70.1 % of all reported syntheses (Fig. S4). The t-SNE analysis reveals distinct property domains for these two methodologies (Fig. 3a). Impregnation studies focus on metal loading and specific surface area, with high-surface-area substrates showing significant positive correlation with metal loading efficiency (Pearson's $r = 0.86$, $p < 0.05$) (Fig. 3b, Fig. S5). Pyrolysis studies emphasize coordination environment parameters, with nitrogen-bearing precursors and NH_3 atmosphere favoring M-N_x coordination structures, while temperature governs the nitrogen content during synthesis (Fig. 3b, Fig. S6–8). Additionally, a systematic comparison of emerging techniques reveals distinct trade-offs in metal loading control, active site uniformity, and stability (Fig. S9, detailed in Supplementary Note S2).

Translating catalyst properties into effective Fenton performance requires navigating conditional dependencies between design variables and application contexts. Our analysis reveals several generalizable principles. The coordination environment emerges as a master variable that simultaneously governs multiple performance dimensions. High-coordination M-N_x sites ($x \geq 4$) anchored in matrices maintain structural stability, with approximately 73 % of studies reporting broad pH tolerance (pH 3–9 in 83 % of studies, Fig. S10). In contrast, oxygen-containing configurations show diminished stability due to M-O bond lability under varying protonation states. This pH dependence further interacts with activator choice. Analysis of reported studies reveals that approximately 87 % of PMS-activated SACs retain high catalytic activity under neutral-to-alkaline conditions, compared to only ~43 % for H_2O_2 -activated systems (Fig. 3c). This disparity can be attributed to the fact that radical generation during PMS activation is less susceptible to pH-induced changes in metal speciation than H_2O_2 activation pathways (Koiki et al., 2023).

Optimal catalyst configuration shifts substantially when pollutant molecular structure is considered. Among the pollutants studied in Fenton-like systems, small-molecule azo dyes (26 % of studies in our corpus) and phenolic compounds (34 %) are the most frequently investigated targets (Fig. 3d). These small molecules exhibit the highest reaction rate constants and achieve uniformly high mineralization rates (>85 %) across diverse coordination configurations (Fig. 3d, Fig. S11). For multi-ring compounds such as tetracycline antibiotics, mixed M-N_xO_y configurations significantly outperform pure M-N_x (mineralization >80 % vs <70 %, Fig. S12). One plausible explanation is that oxygen coordination sites can serve as Lewis acid centers, potentially enhancing adsorption through interactions with carbonyl and hydroxyl functional groups and enabling adsorption-coupled oxidation pathways that favor the degradation of sterically hindered molecules (Cheng et al., 2023). Halogenated compounds favor electron-deficient coordination environments with lower coordination numbers that promote inner-sphere electron transfer for reductive dehalogenation (Fig. S13) (Min et al., 2023). These pollutant-specific requirements underscore that optimal catalyst design must account for target pollutant characteristics.

The activator system selection introduces strong conditional dependencies on optimal catalyst configuration. PMS-activated systems achieve approximately 4-fold higher average rate constants compared to H_2O_2 under comparable conditions (Fig. 3e), consistent with the higher oxidation potential and efficient sulfate radical generation. This performance advantage, however, comes with economic trade-offs: PMS

costs substantially more than H_2O_2 , which explains why H_2O_2 -based systems predominate in pilot-scale demonstrations (88 % of cases in our corpus, Fig. S14) despite their lower intrinsic reactivity (Liu et al., 2024). The choice between activators thus depends on application context: PMS is favored for recalcitrant pollutants where enhanced oxidizing power justifies higher costs, while H_2O_2 remains preferred for large-scale operations prioritizing economic feasibility. More fundamentally, our analysis revealed that optimal coordination environments differ substantially between activator systems (Fig. 3f). For example, Fe-based SACs show preference for lower coordination numbers when paired with PMS but higher coordination with H_2O_2 (Fig. S15). This divergence reflects the distinct activation mechanisms of the two oxidants. PMS activation is generally facilitated by coordinatively unsaturated sites that enhance adsorption and electron transfer, whereas H_2O_2 activation through Fe(II)/Fe(III) redox cycling is favored by higher coordination environments.

These findings collectively reveal that Fenton-like SAC design constitutes a high-dimensional optimization problem characterized by strong conditional dependencies among design variables. The optimal metal center, coordination environment, and synthesis route shift depending on the target pollutant characteristics, pH operating window, and activator selection. This recognition motivates the development of a ML-assisted decision support platform capable of learning and applying these conditional structure-performance relationships.

2.4. Decision-support platform for rational catalyst design

Building upon the structured knowledge base and the recognition that SAC design involves navigating a complex, conditional design space, we developed an interactive decision-support platform that integrates ML-based performance prediction with synthesis recommendation. The platform enables researchers to identify optimal catalyst configurations for specific environmental scenarios and obtain corresponding preparation protocols, effectively translating the extracted knowledge into actionable design guidance.

The predictive core of the platform is a ML model designed to predict catalytic performance under user-specified application conditions. A key design principle of this model is the joint encoding of catalyst properties and environmental parameters as input features, enabling prediction of how specific catalyst configurations perform under specific application scenarios rather than learning a single universal optimum. The training dataset comprises 912 SAC systems extracted from our knowledge baseline, where each data entry represents a unique combination of catalyst configuration and application condition with its experimentally measured performance. The input feature vector for each entry consists of two components: catalyst descriptors including central metal, coordination number, coordinating atom, support type, metal loading, oxidation state, specific surface area, and binding energy; and environmental parameters including pollutant category and pH range. Categorical variables (metal type, coordinating atom, support type, pollutant category, pH range) were one-hot encoded, while numerical variables were standardized, yielding 22 input features in total. Separate models were trained for PMS and H_2O_2 systems to account for their distinct activation mechanisms. The prediction target is the standardized reaction rate constant (k , min^{-1}). Table S2 and Fig. S16 summarize the inputs and distribution of initial datasets, respectively, showing that most variables are well-distributed within normalized values and suitable for modeling. Analysis of Pearson's correlation coefficient shows that there is low linear correlation between variables (Fig. S17), which confirms the minimal multicollinearity among features for the ML models training.

Several ML models including Random Forest, XGBoost, LightGBM, and LASSO regression were selected to train the initial dataset. These models were chosen to explore both linear relationships (primarily via LASSO) and non-linear interactions (via the tree-based Random Forest, XGBoost, and LightGBM) anticipated from the mixed numerical and one-

hot encoded categorical input features. Prior to model training, we implemented an 80/20 training-test set split stratified by activator type and used 10-fold cross-validation to prevent data leakage, ensuring the generalizability of the model. The detailed ML methodology is described in **Experimental Procedures**. R^2 , MSE, and MAE were employed to assess the ML models. **Fig. S18-S19** presents the comprehensive comparison of the model evaluation outcomes. The XGBoost model demonstrated superior performance, achieving the highest R^2_{test} of 0.91 and 0.87 for the PMS and H_2O_2 systems among the four trained models, indicating its ability to explain a larger proportion of the variance and establish a stronger linear relationship with the dependent variable. Additionally, the exhibited lowest MAE and MSE values of XGBoost model confirm its accuracy (**Fig. 4a**). SHAP analysis was employed to quantify feature contributions to model predictions (**Fig. 4b**). The analysis revealed that central metal atom and coordination number rank as the most influential catalyst descriptors, while activator type and pollutant category emerge as the dominant environmental parameters. Further analysis of SHAP values across coordination numbers shows that PMS systems favor lower coordination numbers while H_2O_2 systems benefit from higher coordination (**Fig. 4c**), consistent with our earlier statistical findings.

The trained model enables the platform to answer the core question: given a specific wastewater treatment scenario, what catalyst configuration will achieve optimal performance? The platform operates through

a query interface where researchers sequentially complete four input fields to define their application requirements (**Fig. 4d**): (1) target pollutant category (phenolic compounds, azo dyes, antibiotics, or halogenated organics), (2) operating pH range (acidic, neutral, or alkaline), (3) activator system (PMS or H_2O_2), and (4) optional metal element constraints based on cost or availability considerations. These selections define the environmental condition inputs that will be held constant during the subsequent optimization process. Upon submission, the platform backend executes a systematic screening process to identify optimal catalyst configurations (**Fig. 4d**). The user-specified environmental parameters (pollutant type, pH range, activator) are encoded and fixed as the environmental portion of the model input vector. The platform then generates a candidate library by enumerating all feasible combinations of unconstrained catalyst design variables, including metal center, coordination number, coordinating atoms, and support type. For each candidate, the platform constructs a complete input vector by combining the enumerated catalyst descriptors with the fixed environmental parameters, then feeds this vector to the trained XGBoost model to predict the expected k value (min^{-1}). After evaluating all candidates, the platform ranks them by predicted performance and presents the top configurations along with predicted k values and confidence estimates (mean $\pm$ one standard deviation across the 10 cross-validation fold models; calibration details in **Supplementary Note S5**).

Each recommendation is further classified into one of three

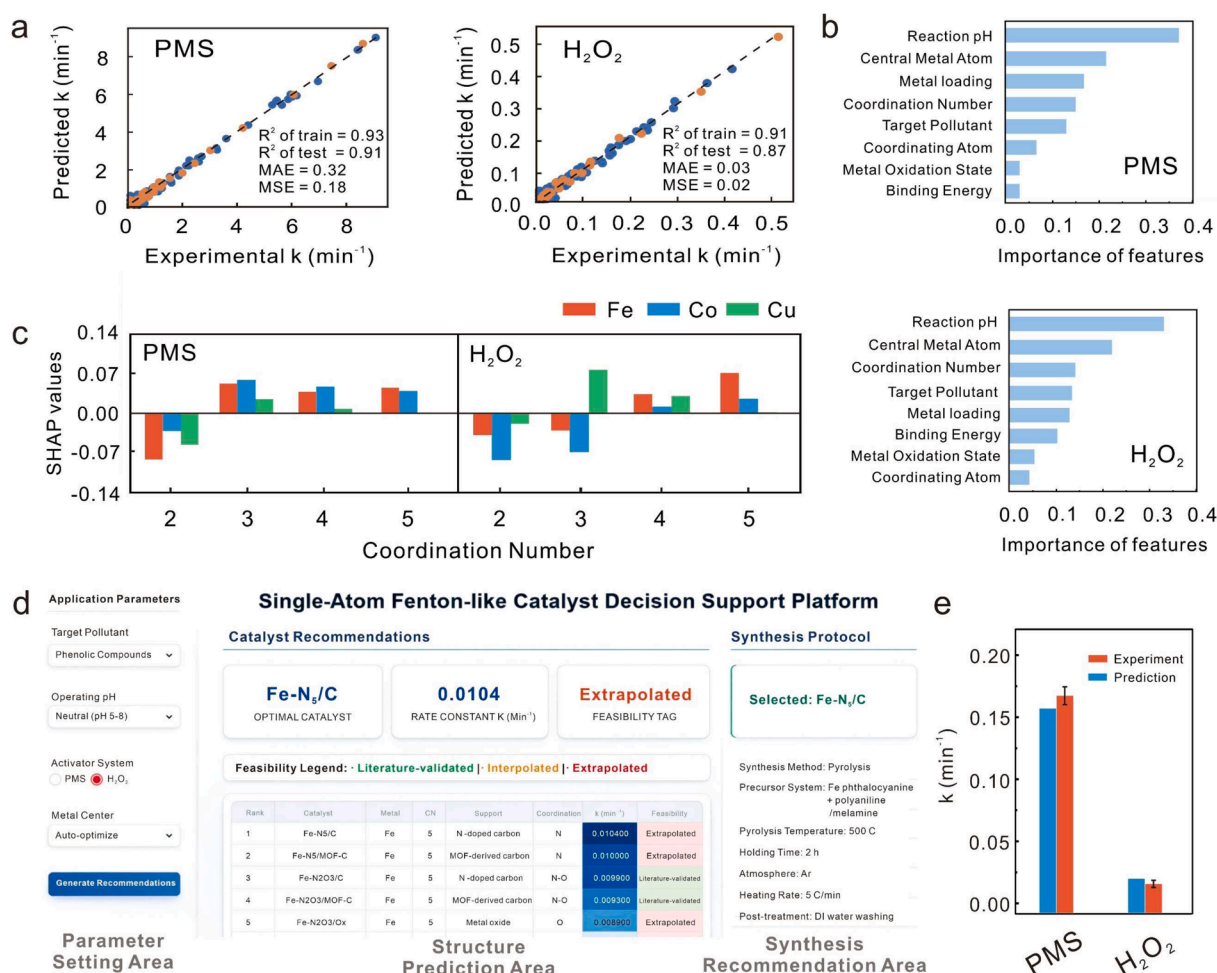


Fig. 4. ML-driven decision support platform for rational catalyst design. (a) Parity plots comparing predicted versus experimental reaction rate constants for PMS and H_2O_2 systems, demonstrating model accuracy. (b) SHAP feature importance analysis identifying key descriptors governing catalytic performance in PMS and H_2O_2 systems. (c) SHAP value distributions for different coordination numbers across Fe, Co, and Cu metal centers, revealing activator-dependent structure-performance relationships. (d) Interactive decision-support platform interface featuring parameter setting, structure prediction, and synthesis recommendation modules. (e) Experimental validation confirming platform-recommended catalysts achieve predicted performance under realistic conditions.

feasibility tiers: “literature-validated” when the recommended configuration has direct experimental precedent in the knowledge baseline under comparable conditions, “interpolated” when the configuration falls within the training data feature space but lacks an exact match, and “extrapolated” when the configuration involves feature combinations outside the training data distribution. This classification enables users to assess the risk level of each recommendation before committing experimental resources. It should be noted that the predictive reliability of the platform is inherently linked to data coverage in the knowledge baseline. Categories well-represented in the corpus (e.g., Fe, Co, and Cu as central metals) yield narrower confidence intervals, whereas predictions involving underrepresented categories carry greater uncertainty and should be treated as hypothesis-generating rather than definitive. Readers are encouraged to consult the data distribution figures presented in this work to assess coverage for specific catalyst-condition combinations before acting on platform recommendations.

Following configuration recommendation, the platform queries the synthesis-property knowledge base to retrieve preparation protocols for the top-ranked catalysts (Fig. 4d). Additionally, the platform traces back to the original literature to provide experimentally validated protocols that have successfully yielded the recommended structures. Given that pyrolysis and impregnation represent the two dominant synthesis strategies for SACs in our corpus, the platform organizes its recommendations around these established approaches. For each recommended configuration, the platform retrieves representative synthesis protocols from multiple source publications, presenting key parameters including precursor types, treatment temperatures, atmospheres, and durations. This literature-anchored approach acknowledges that multiple valid routes may exist for achieving similar structural outcomes, and provides users with documented starting points that can be adapted to their specific laboratory conditions. The platform displays source references alongside each protocol, enabling users to consult the original publications for detailed procedures and to select routes most compatible with their available resources. It should be noted that reaction parameters such as temperature, solid-liquid ratio, and activator dosage are standardized during prediction to ensure fair comparison across catalyst configurations. An interactive demonstration with the complete source code is available at https://github.com/HaoyangFu/fenton_sac_platform.

In practice, when a user specifies phenol degradation using PMS activation at neutral pH without metal constraints, the platform encodes these environmental conditions (pollutant = phenolic, activator = PMS, pH = neutral, central atom = Fe) and fixes them in the input vector (Fig. S20). It then evaluates all catalyst combinations by predicting their performance under these specific conditions. The model, having learned from training examples that Fe-N₃ configurations exhibit superior *k* values when paired with PMS activation for phenolic degradation, ranks these configurations highest and recommends nitrogen-doped carbon as the support material based on performance correlations in the training data. The platform then retrieves pyrolysis-based synthesis protocols with specific temperature, atmosphere, and duration parameters traced from source publications. When the same user changes only the activator to H₂O₂, the platform re-evaluates all candidates under the new environmental encoding and identifies Fe-N₅ as optimal, retrieving impregnation-based protocols using iron salt loading followed by thermal treatment. Each protocol is accompanied by its literature reference, enabling users to consult original publications and adapt procedures to their laboratory conditions.

Briefly, upon receiving user-defined environmental inputs (pollutant category, pH range, activator type, and optional metal constraints), the platform encodes these parameters and systematically evaluates all feasible catalyst configurations by feeding each candidate descriptor vector into the trained XGBoost model to predict the expected reaction rate constant. Candidates are then ranked by predicted performance, and corresponding synthesis protocols are retrieved from the knowledge base for the top-ranked configurations. To validate the reliability of this

decision-support system, we prepared Fe SACs following platform-recommended protocols and characterized their structures using X-ray Absorption Spectroscopy (XAS). Fitting analysis confirmed that the pyrolysis route yielded Fe-N₃ coordination while the impregnation route produced Fe-N₅ coordination (Table S3), verifying that the recommended synthesis protocols successfully achieved the target structures. Catalytic evaluation demonstrated that Fe-N₃ achieved *k* values of $0.17 \pm 0.01 \text{ min}^{-1}$ in PMS systems, while Fe-N₅ achieved $0.03 \pm 0.005 \text{ min}^{-1}$ in H₂O₂ systems (Fig. 4e). These results align precisely with platform predictions, demonstrating that the ML model successfully learned the conditional structure-performance relationships and the platform reliably translates this knowledge into actionable recommendations. Both catalysts maintained satisfactory degradation performance over five consecutive cycles (Fig. S21) and retained robust activity in a simulated complex water matrix containing common competing species (NaCl, NaHCO₃, Na₂SO₄, and humic acid) (Fig. S22), suggesting that the platform-recommended catalysts hold promises for practical wastewater treatment applications. To further assess the generalizability of the platform, we queried a different application scenario (azo dye, acidic pH). The platform again recommended Fe-N₃/C and Fe-N₅/C for the PMS and H₂O₂ systems, respectively, and supplementary validation using methylene blue confirmed close agreement between predicted and measured *k* values (Fig. S23). Through this comprehensive framework integrating knowledge extraction, ML-driven prediction, and synthesis recommendation, we establish a data-driven paradigm that transforms catalyst design from experience-dependent trial-and-error into systematic, scenario-specific decision-making for wastewater treatment.

3. Conclusion

In summary, we developed an LLM-driven framework that enables rapid, high-fidelity extraction of scientific knowledge from over 1000 domain publications through optimized few-shot CoT prompting, a task that would otherwise require months of manual effort. The resulting knowledge baseline reveals quantitative relationships between synthesis parameters, structural descriptors, and catalytic performance. Building on this foundation, we established a ML-assisted decision-support platform that predicts catalyst performance under user-specified conditions and recommends optimal configurations with corresponding synthesis protocols. Experimental validation confirms that platform-recommended catalysts achieve predicted performance, establishing a closed loop from literature mining to rational catalyst design. In its current form, the platform operates within the established literature design space, guiding researchers toward optimal configurations for specific application scenarios. We expect this methodology to serve as a transferable paradigm for intelligent decision-making across broader wastewater treatment systems.

CRedit authorship contribution statement

Haoyang Fu: Data curation, Conceptualization. **Qingze Chen:** Project administration, Methodology, Funding acquisition. **Chenliu Tang:** Funding acquisition, Formal analysis, Data curation. **Zhongyi Deng:** Data curation, Conceptualization. **Jingxiang Ng:** Software, Resources. **Junbo Niu:** Methodology, Investigation. **Runliang Zhu:** Writing – review & editing, Supervision. **Shuzhou Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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